Association of dietary fat, vegetables and antioxidant micronutrients with skin ageing in Japanese women.

Daily diet may have implications for skin ageing. However, data on the relationship between diet and the parameters of skin conditions are scarce. The present study aimed to examine the associations of biophysical properties of the skin of women with intakes of fats and antioxidant micronutrients as well as food groups as sources of these nutrients. In a cross-sectional study, we measured the hydration, surface lipids and elasticity of the skin of 716 Japanese women using non-invasive techniques. The extent of facial wrinkles in the crow's-foot area was determined by observation using the Daniell scale. Each subject's usual diet was determined with the use of a validated FFQ. After controlling for covariates including age, smoking status, BMI and lifetime sun exposure, the results showed that higher intakes of total fat, saturated fat and monounsaturated fat were significantly associated with increased skin elasticity. A higher intake of green and yellow vegetables was significantly associated with a decreased Daniell wrinkling score. Intake of saturated fat was significantly inversely associated with the Daniell wrinkling score after additional adjustment for green and yellow vegetable intake. Further studies with more accurate measurement methods are needed to investigate the role of daily diet in skin ageing.